

1. Title

Automated Quality Control (AQC) System for Semiconductors and Motherboard Manufacturing

2. Problem Statement

TechManufacturing Inc. is currently facing substantial operational challenges due to increasing quality control costs and high rates of product recalls. The facility produces approximately 10,000 units daily, including intricate Printed Circuit Boards (PCBs) with mounted chips and components that require meticulous inspection for defects such as scratches, component misalignments, and soldering issues. The current reliance on manual inspection is not only time-consuming and expensive but also prone to human error, which directly impacts the brand's reliability and bottom line.

3. Description

The proposed solution implements an AI-native engineering framework to deploy a high-speed Computer Vision (CV) pipeline at the terminus of the production line.

- **Model Architecture:** Utilization of the YOLO (You Only Look Once) model for real-time object detection and defect classification.
- **Development Workflow:** Adoption of an AI-native approach using VS Code with Agentic mode and GitHub Copilot to accelerate the creation of robust inference logic and data preprocessing pipelines.
- **Deployment Infrastructure:** The system will run on high-performance compute servers equipped with dedicated GPUs (e.g., NVIDIA A100 or H100) to handle the significant hardware demands of real-time YOLO inference.
- **Integration:** The software will be containerized using Docker to ensure consistent performance across the manufacturing floor's local server environment.

4. Justification of the Solution

The selection of these specific methods and technologies is based on the following technical advantages:

- **Efficiency of YOLO:** The YOLO model is specifically chosen for its ability to perform high-speed, real-time detection, which is essential for a production line handling 10,000 units per day.
- **AI-Native Productivity:** Using Agentic workflows in VS Code allows engineers to rapidly iterate on complex AI components, ensuring that the logic for identifying subtle defects is both rigorous and easily maintainable.
- **Infrastructure Reliability:** Leveraging on-premise GPU-accelerated servers ensures low-latency processing and high data security, keeping sensitive manufacturing data within a closed-server architecture.
- **Scalability via Containerization:** Utilizing Docker and Kubernetes ensures that as production volume increases, the AI services can be scaled horizontally to meet demand without significant downtime.